Data quality notes: BUSM1042 Assignment 3 hotel bookings dataset

Prepared for the group report; every count is linked to BUSM1042_A3_Group_Clean_Data.xlsx

1. How the data was checked

The team worked on a copy of the supplied file and kept the original unchanged. Each column was checked against the variable definitions in the Data in Brief paper that published the dataset (Antonio, de Almeida & Nunes 2019): what it means, what kind of values it holds and anything that looks odd. Quality was judged by fitness for use, so a problem matters only if it could change an answer to one of our five business questions (Vidgen, Kirshner & Tan 2019, p. 58). Suspect values were flagged rather than deleted or guessed, because deleting cases can bias results (Vidgen, Kirshner & Tan 2019, p. 63). As the scenario requires, every booking is treated as an Australian tourist's, so the country column is never used to cut the data.

2. Data quality issues and treatment

The table lists the 20 issues found, in the order of the workbook's Data_Quality_Log sheet. No row was deleted: problems are handled with flags, recoded copies of columns and sensitivity checks.

Table X: Data quality checks and treatment

Quality dimension	What was found	Rows affected	What was done and why	Effect on results	Questions affected
DQ01 Consistency	is_canceled and is_repeated_guest are coded Yes/No; the dictionary and the paper say 0/1	119,390	Recoded to 1/0 in new flag columns (cancel_flag, repeat_flag). The average of a 1/0 flag is the rate.	None; coding only	Q1–Q5
DQ02 Consistency	arrival_date holds full dates, but the dictionary says "Year of arrival"	119,390	Year and month derived from the date (arrival_year, arrival_month). Lets the data be analysed by month.	None	Q1
DQ03 Validity	96 rows hold the impossible text date "29 February 2023" (most likely the original 29 February 2016 after the years were shifted)	96	Flagged (date_valid = 0); left out of date-based analysis only. The true date cannot be checked, so no new date is guessed.	February arrivals are undercounted by 96; no other result changes	Q1

Quality dimension	What was found	Rows affected	What was done and why	Effect on results	Questions affected
DQ04 Completeness	children stored as the text "NA"	4	Converted to a blank; flagged invalid (flag_invalid). The value cannot be checked.	Included in the sensitivity test. See Sensitivity sheet: excluding all 428 flagged rows changes the overall rate by +0.03 points	All (flag test only)
DQ05 Completeness	country stored as the text "NULL"	488	Converted to a blank, labelled "Unknown" (country_clean). Country is never used to filter (scenario rule).	None	None (country not used)
DQ06 Consistency	Both "CN" (1,279) and "CHN" (999) appear; the paper uses 3-letter ISO codes	1,279	Flagged only, NOT merged. It is probably the same country, but this cannot be verified, and country is not used.	None	None (country not used)
DQ07 Completeness	agent "NULL" (16,340) and company "NULL" (112,593)	16,340 / 112,593	Converted to blanks and recoded "None" (agent_clean, company_clean). The paper says NULL means not applicable (p. 45); an optional field left empty is still complete (Vidgen, Kirshner & Tan 2019, p. 57).	None	None
DQ08 Completeness	market_segment "Undefined" (2) or distribution_channel "Undefined" (5)	5	Flagged (flag_undefined), kept. The true value cannot be checked.	Excluding them moves the overall rate by less than 0.01 points	Q3, Q4
DQ09 Consistency	meal "Undefined" (1,169); the paper defines Undefined/SC as no meal package	1,169	meal_clean maps Undefined → SC (stated assumption); the original meal column is kept. The paper documents the meaning (Table 1, p. 44).	SC rate goes from 37.2% (10,650 bookings) to 36.0% (11,819)	None (meal not used)
DQ10 Validity	No guests at all (adults + children + babies = 0)	180	Flagged invalid (flag_invalid), kept. A booking with no guests cannot be right.	See Sensitivity sheet: excluding all 428 flagged rows changes the overall rate by +0.03 points	All (flag test only)
DQ11 Validity	No adults but children or babies present	223	Flagged outlier (flag_outlier), kept. Unusual, but possible (for example, a booking error or a minors' group).	See Sensitivity sheet: excluding all 428 flagged rows changes the overall rate by +0.03 points	All (flag test only)
DQ12 Validity	Extreme party sizes: adults ≥ 10 (13 rows), children 10 (1), babies 9 or 10 (2)	16	Flagged outlier (flag_outlier), kept. Possibly group entries or typing errors; cannot be checked.	See Sensitivity sheet: excluding all 428 flagged rows changes the overall rate by +0.03 points	All (flag test only)
DQ13 Validity	Average_Daily_Rate of −6.38 (1) and 5,400 (1, about 57 times the median of 94.575)	2	Flagged invalid (flag_invalid), kept. A negative price is impossible; 5,400 is almost certainly an entry error.	See Sensitivity sheet: excluding all 428 flagged rows changes the overall rate by +0.03 points	All (flag test only)

Quality dimension	What was found	Rows affected	What was done and why	Effect on results	Questions affected
DQ14 Validity	Average_Daily_Rate = 0 (680 of them Complementary)	1,959	Flagged (flag_zero_rate), kept. Many are complimentary stays or stays of zero nights; these are not errors.	Exclude them only for revenue or price analysis	None (no price question)
DQ15 Validity	Zero nights (weekend + week nights = 0); all have a rate of 0	715	Flagged (flag_zero_nights), kept. Possibly day use or same-day changes; cannot be checked.	Excluding them moves the overall rate from 37.0% to 37.2%	All (very small effect)
DQ16 Validity	required_car_parking_spaces = 8	2	Flagged outlier (flag_outlier). Implausible for one booking.	See Sensitivity sheet: excluding all 428 flagged rows changes the overall rate by +0.03 points	All (flag test only)
DQ17 Uniqueness	Rows identical on all 28 variables but with different booking IDs	32,239	Kept, flagged (dup_flag). 78.5% are Groups or Offline TA/TO bookings (against 36.9% of all bookings), so most likely separate rooms in multi-room bookings.	Removing them would cut the overall rate from 37.0% to 27.3%; report patterns, and check each key result with dup_flag = 0	Q1–Q5
DQ18 Consistency	is_canceled = Yes covers 43,017 cancellations and 1,207 no-shows; it agrees with reservation_status in every row	44,224	Kept: "cancellation" includes no-shows (state this in the report). That is the dataset's own definition.	Excluding no-shows lowers the overall rate from 37.0% to 36.0%	Q1–Q5
DQ19 Timeliness	Arrivals cover 1 July 2022 to 31 August 2024, so 2022 and 2024 are part-years (21,996 / 56,611 / 40,687 valid-date rows)	119,294	Note only. Comparing yearly totals would be misleading.	Compare rates or months, not yearly booking totals	Q1
DQ20 Accuracy, fitness for use (Vidgen, Kirshner & Tan 2019, p. 58)	Some fields look as if they were filled in at or after check-in. 0 of 7,416 bookings with a parking space were cancelled, and only 5.4% of bookings with a changed room type (14,917 rows). Cancelled guests never arrive, so they never get a parking space or a room change	7,416 / 14,917	Kept; marked "use with caution". These fields partly record the outcome rather than cause it; the paper warns that guests often change such details at check-in (p. 45).	Do NOT present parking or room changes as reasons guests cancel	Any using parking or room changes

Source: Group analysis of the supplied hotel bookings dataset (n = 119,390); quality dimensions from Vidgen, Kirshner and Tan (2019, pp. 57–58).

3. How much could these choices change our results?

Excluding all 428 flagged rows (invalid, undefined or outlier values) moves the overall cancellation rate from 37.04% to 37.07%, a change of only +0.03 points, so these flags do not change any overall conclusion. Two choices do matter. Removing the 32,239 duplicate rows would cut the rate from 37.0% to 27.3%, and counting no-shows as not cancelled would lower it to 36.0%. Every question should therefore state that cancellations include no-shows and check its main pattern without duplicates (dup_flag = 0). Inside single tables, a few small

categories move more, for example customer type Group, which moves by 2.19 points, and teammates using those variables should say so.

Table Y: How much each decision changes the overall cancellation rate

Test	Bookings used	Overall cancellation rate	Change (points)
All bookings (baseline)	119,390	37.04%	baseline
Excluding all flagged rows (flag_any_quality)	118,962	37.07%	+0.03
Excluding invalid and undefined rows only	119,203	37.07%	+0.03
Excluding zero-night stays	118,675	37.24%	+0.19
Excluding duplicates (dup_flag = 1)	87,151	27.29%	-9.76
Counting only 'Canceled' (no-shows not counted)	119,390	36.03%	-1.01

Source: Sensitivity sheet of BUSM1042_A3_Group_Clean_Data.xlsx. Points = percentage points.

4. Notes for each business question

Q1 (size of the problem; City vs Resort; over time). Use date_valid = 1 for any monthly analysis, because 96 bookings have an impossible date. Arrivals run from 1 July 2022 to 31 August 2024, so 2022 and 2024 are part-years: compare rates or months, not yearly totals. Never split or filter by country. City Hotel bookings were cancelled 41.7% of the time, against 27.8% for Resort Hotel.

Q2 (lead time). Use the six lead-time bands. The Over 365 band is small (2.6% of bookings) and 59.2% of it is non-refundable, so its high rate (67.7%) partly reflects deposit policy rather than time alone.

Q3 (no-deposit bookings; channels and segments). The 5 bookings with an Undefined segment or channel are flagged. Aviation (237 bookings) and Complementary (743) are small segments, and GDS has only 193 bookings, so report their rates with care.

Q4 (deposit type × segment). Non Refund bookings were cancelled 99.4% of the time, against 28.4% for No Deposit. Refundable has only 162 bookings, so its rates (22.2% overall) are unreliable once it is split by segment; say so.

Q5 (booking history). Values of previous_cancellations from 11 to 26 (180 rows) are probably repeated group records: 84.4% of them are Groups or Offline TA/TO bookings. Treat them as one '2+' band (prev_canc_band). Repeat guests cancelled 14.5% of their bookings, against 37.8% for other guests.

For everyone. Parking and room changes are 'use with caution' (DQ20). 0 of 7,416 bookings with a parking space were cancelled, and only 5.4% of bookings with a changed room type, because these details are often set at check-in. Do not present them as reasons guests cancel.

5. What stands out (to help write the business questions)

Each bullet compares the highest and lowest cancellation rate within one variable, using only categories with 30 or more bookings; bullets run from the largest gap to the smallest. Rates come from the OneVar_Summary sheet.

- Deposit type: Non Refund 99.4% vs Refundable 22.2% (a gap of 77.1 points); No Deposit, the largest group, 28.4%; check whether this is real behaviour or a recording practice (report Section 6.4).
- Previous cancellations: guests with one previous cancellation 94.4% vs guests with no previous cancellation 33.9% (a gap of 60.5 points).
- Lead time: Over 365 days 67.7% vs 0–7 days 9.6% (a gap of 58.0 points).
- Market segment: Groups 61.1% vs Complementary 13.1% (a gap of 48.0 points); unreliable (n < 30): Undefined (2 bookings, 100.0%).
- Parking spaces: no parking space 39.5% vs one or more parking spaces 0.0% (a gap of 39.5 points); use with caution, as this may be recorded at check-in (DQ20).
- Total nights: 2–3 nights 43.0% vs 0 nights 4.9% (a gap of 38.1 points).
- Room type changed: same room type 41.6% vs changed room type 5.4% (a gap of 36.2 points); use with caution, as this may be recorded at check-in (DQ20).
- Previous bookings not cancelled: guests with no earlier booking that was kept 38.0% vs guests with one or more earlier bookings that were kept 5.5% (a gap of 32.5 points).
- Special requests: no special requests 47.7% vs three or more requests 16.8% (a gap of 30.9 points).
- Customer type: Transient 40.7% vs Group 10.2% (a gap of 30.5 points).
- Average daily rate: a rate of 50.01–100 38.9% vs a rate of 0 or less 10.9% (a gap of 28.1 points).
- Days on the waiting list: one or more days waiting 63.8% vs no days on the waiting list 36.2% (a gap of 27.6 points).
- Booking changes: no booking changes 40.9% vs one change 14.2% (a gap of 26.6 points).
- Meal (Undefined counted as SC): FB 59.9% vs HB 34.5% (a gap of 25.4 points).
- Distribution channel: TA/TO 41.0% vs Direct 17.5% (a gap of 23.6 points); unreliable (n < 30): Undefined (5 bookings, 80.0%).
- Repeat guest: other guests 37.8% vs repeat guests 14.5% (a gap of 23.3 points).
- Hotel: City Hotel 41.7% vs Resort Hotel 27.8% (a gap of 14.0 points).

- Reserved room type: type H 40.8% vs type E 29.3% (a gap of 11.5 points); use with caution, as this may be recorded at check-in (DQ20); unreliable (n < 30): L (6 bookings, 33.3%), P (12 bookings, 100.0%).
- Arrival month (valid dates only): June 41.5% vs January 30.5% (a gap of 11.0 points).
- Arrival year: 2024 38.7% vs 2023 35.9% (a gap of 2.8 points); 2022 and 2024 are part-years, so compare rates only.
- Children or babies: adults only 37.2% vs with children or babies 34.9% (a gap of 2.3 points).

6. Limitations

- The file has no cancellation date (the paper's ReservationStatusDate is not included), so the notice hotels receive before a cancellation cannot be measured.
- The patterns are associations, not causes: a variable linked with a higher cancellation rate does not on its own make guests cancel.
- Some fields may be recorded at check-in, such as parking spaces and room changes (DQ20), so they partly describe the outcome rather than explain it.

7. Reference list

Antonio, N, de Almeida, A & Nunes, L 2019, 'Hotel booking demand datasets', *Data in Brief*, vol. 22, pp. 41–49, viewed 3 November 2026, <https://doi.org/10.1016/j.dib.2018.11.126>.

Vidgen, R, Kirshner, S & Tan, F 2019, *Business analytics: a management approach*, Red Globe Press, London.

Note for the group: if the report also cites other Antonio, de Almeida & Nunes 2019 papers, label them 2019a, 2019b and 2019c consistently. In Mark's sections the Data in Brief paper is 2019c.